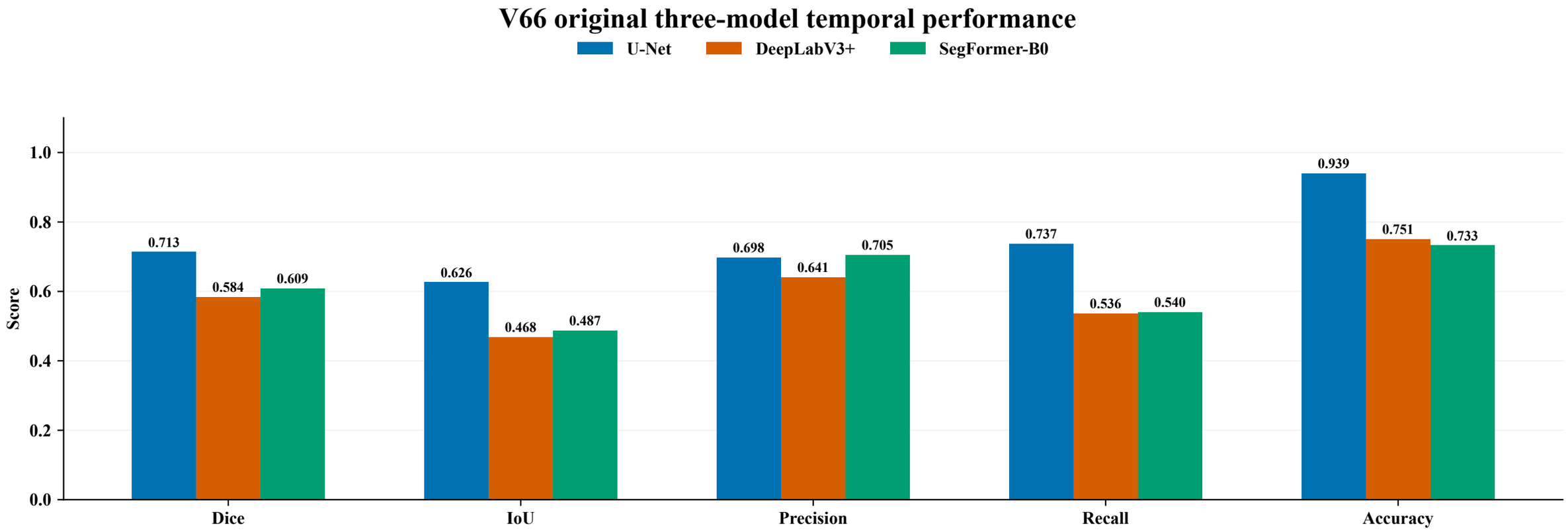

Model	Macro Dice	Macro IoU	Macro precision	Macro recall	Root accuracy
U-Net	0.7132	0.6264	0.6978	0.7368	0.9393
DeepLabV3+	0.5837	0.4684	0.6408	0.5364	0.7510
SegFormer-B0	0.6089	0.4868	0.7053	0.5400	0.7327

Root-only manual-reference agreement; T2-T4 pooled. First four metrics are class macro averages; missed roots are retained.